

KRIVA PALANKA, North Macedonia

WMO: 134930

Lat: 42.200N

Lon: 22.333E

Elev: 2283

StdP: 13.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	17.1	4.7	7.6	14.8	9.9	9.9	20.3	17.7	41.1	15.7	34.9	9.2	50	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.7	91.2	67.6	88.2	66.4	85.2	65.5	69.3	86.6	68.0	84.6	66.6	82.2	6.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	96.5	75.4	62.3	91.4	73.6	61.0	87.1	72.3	34.8	87.3	33.6	84.5	32.5	82.5	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.6	14.0	12.8	DB	4.1	95.0	5.9	4.0	-0.1	97.8	-3.6	100.2	-6.9	102.4	-11.2	105.3	
			WB	4.0	72.4	5.3	1.4	0.2	73.3	-2.9	74.1	-5.8	74.9	-9.7	75.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	32.9	37.2	43.8	51.6	59.0	65.5	69.9	70.8	62.5	53.7	45.4	35.9
	DBStd	14.46	8.22	7.96	6.73	6.33	5.71	5.81	5.13	4.65	6.42	7.17	7.22	7.73
	HDD50	1814	531	360	207	52	4	0	0	0	2	49	169	440
	HDD65	5080	996	778	656	403	197	64	15	8	120	354	587	902
	CDD50	2704	0	2	16	99	284	466	616	646	377	163	32	3
	CDD65	495	0	0	0	0	12	80	166	190	45	2	0	0
	CDH74	5964	0	0	0	25	220	988	1993	2118	568	50	2	0
	CDH80	2128	0	0	0	1	35	282	792	850	163	5	0	0
Wind	WSAvg	7.5	6.9	7.9	7.8	7.8	7.5	7.3	7.6	7.9	7.7	7.7	7.2	7.0
Precipitation	PrecAvg	23.7	1.5	1.4	1.7	2.3	2.4	2.4	1.8	1.8	2.0	2.3	2.1	2.1
	PrecMax	37.5	4.2	4.1	4.1	6.1	4.9	5.0	5.4	5.1	5.7	5.2	3.8	5.0
	PrecMin	13.3	0.1	0.1	0.0	0.2	1.0	0.5	0.0	0.3	0.4	0.1	0.4	0.0
	PrecStd	5.2	1.1	0.9	0.9	1.3	1.0	1.2	1.1	1.1	1.4	1.5	0.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	62.8	69.0	77.4	84.2	90.8	95.3	95.4	89.4	79.2	69.2	59.1
		MCWB	46.3	48.7	52.2	57.3	63.2	68.0	68.9	68.4	65.1	61.4	55.2	50.8
	2%	DB	51.7	57.6	64.5	72.2	79.6	86.7	91.4	91.3	84.8	74.7	64.3	54.1
		MCWB	43.9	46.5	49.7	55.8	61.5	66.8	67.8	67.7	63.7	58.7	53.6	47.0
	5%	DB	48.1	53.5	60.9	68.1	76.1	83.5	88.0	88.2	80.7	71.0	60.7	50.2
		MCWB	42.1	44.7	47.8	53.8	60.2	65.3	66.8	66.1	62.2	57.2	51.8	44.9
	10%	DB	44.3	49.7	56.7	64.5	72.9	80.3	84.7	84.9	76.5	66.9	57.0	46.9
		MCWB	40.1	42.6	45.6	51.9	58.8	64.1	65.9	65.2	60.4	55.4	49.9	42.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	50.2	53.1	59.3	65.4	70.6	71.7	71.2	67.0	62.5	57.4	51.2
		MCDB	54.9	60.3	67.0	72.1	80.0	84.9	89.2	89.6	84.2	77.3	66.0	57.1
	2%	WB	44.8	47.6	50.6	56.7	62.9	68.2	69.4	69.2	64.8	60.1	54.6	47.9
		MCDB	50.6	56.1	62.2	69.8	76.7	83.7	86.6	87.6	81.1	71.6	62.4	52.9
	5%	WB	42.6	45.4	48.4	54.8	61.1	66.6	67.9	67.7	63.3	58.1	52.6	45.5
		MCDB	47.1	52.4	58.6	66.7	74.0	80.8	84.2	85.0	77.9	67.8	59.5	49.3
	10%	WB	40.5	43.1	46.6	52.7	59.5	65.0	66.6	66.0	61.6	56.4	50.5	43.1
		MCDB	44.0	49.3	55.7	62.4	70.9	78.0	82.3	81.6	73.7	65.2	56.2	46.3
Mean Daily Temperature Range	5% DB	MDBR	14.3	17.2	19.6	20.0	21.6	23.4	25.4	25.7	22.2	20.3	16.7	14.0
		MCDBR	19.0	23.4	27.0	26.7	27.7	28.8	31.0	30.7	29.6	27.1	21.9	17.3
		MCWBR	12.2	13.7	14.5	12.9	12.3	11.7	11.5	11.3	11.9	13.2	12.4	11.0
	5% WB	MCDBR	16.9	21.6	24.4	24.4	25.7	27.0	27.9	27.9	25.7	23.7	19.8	14.8
		MCWBR	11.4	13.1	13.5	12.2	12.1	11.9	11.4	11.1	11.1	12.2	11.5	10.0
Clear-Sky Solar Irradiance	taub	0.318	0.345	0.388	0.446	0.436	0.421	0.421	0.437	0.412	0.376	0.345	0.319	
	taud	2.375	2.297	2.209	2.059	2.155	2.238	2.237	2.191	2.239	2.328	2.376	2.384	
	Ebn at Noon	260	269	269	261	267	271	270	261	259	254	246	248	
	Edh at Noon	26	33	40	50	47	43	43	44	39	31	26	24	
All-Sky Solar Radiation	RadAvg	515	796	1123	1428	1733	1995	2070	1848	1353	954	612	445	
	RadStd	79	103	133	170	134	163	139	141	111	116	62	73	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-272	N/A	N/A

Nomenclature: See separate page